

3(a)

any four from

- 1 mRNA, attaches to / associates with / AW, ribosome ;
A ribosome reads mRNA
- 2 two codons, exposed / AW ;
A A and P sites / first and second binding sites, (on ribosome)
- 3 tRNA, qualified carries an amino acid to ribosomes / each type carries a specific amino acid / first amino acid is met / tRNA with met / tRNA^{met} ;
ref. to ribosomes can be implied in mp 3 if mp1 gained
- 4 anticodon (on tRNA) binds to codon (on mRNA) ; **I** matches
A complementary / base, pairing between codon and anticodon
A H-bonds form between bases on codon and anticodon
R if anticodon on mRNA
- 5 START (mRNA) codon / (mRNA) AUG first codon ;
- 6 second tRNA with its amino acid binds next to first tRNA / (two) amino acids are held in place close to each other by tRNA binding / AW ;
A *ref. to two tRNAs next to each other*
- 7 (after peptide bond formation) first tRNA detaches / ribosome moves along one codon
or
process repeats / elongation occurs, until, STOP codon / polypeptide chain synthesised (and released from ribosome) ;
- 8 tRNA molecules, reused / leave ribosome to attach to another amino acid ;
must be in context of leaving after peptide bond formation
- 9 AVP ; e.g. peptidyl transferase for peptide bond formation
ref. to, 99 / 100 / 101, codons, qualified
A number of bases in length of mRNA

Question	Answer	Marks
3(b)	<i>any three from similarity</i> peptide bond and disulfide bond are covalent bonds ; <i>differences</i> tertiary structure bonds are between, R groups / side chains, (of different amino acids ; H bond / ionic bond / hydrophobic interaction, versus <u>covalent</u> peptide bond ; A tertiary structure bonds apart from disulfide are not covalent peptide bond, stronger / more thermostable, than, tertiary structure bonds / two named bonds; peptide bond is between, carboxylic acid / <u>COOH</u>, and, amino / <u>NH₂</u>, group (of the adjacent amino acid) ; AVP ; detail of a tertiary structure bond e.g. hydrophobic between non-polar R groups disulfide between sulfur-containing R groups ionic between carboxyl and amino groups of R groups H bond between oxygen on —CO groups and H on either the —OH or —NH groups of R groups	3

Question	Answer	Marks
3(c)(i)	<i>any three from:</i> <i>suggestion that</i> 1 flap region, opens / moves, to let, substrate / polyprotein, enter or flap region, closes round / AW, substrate / polyprotein I flap region is closed 2 induced fit / HIV protease action, involves, change in shape of active site / active site moulding around substrate / AW ; I <i>ref.</i> to enzyme or substrate changing shape <i>if mp1 and mp2 not gained allow one mark if stated that activity of flap region changes shape of active site or that flap region changes shape because it is the active site</i> 3 <i>ref.</i> to better fit / fully complementary / AW ; 4 <i>ref.</i> to formation of, enzyme-substrate complex ; A ESC 5 AVP ; e.g. (induced fit) substrate / polyprotein, not (fully) complementary to (shape of) active site A does not (fully) fit active site / enzyme, returns to original shape on release of product change of shape (to give complementary fit), lowers activation energy / puts strain on bonds / AW	3
3(c)(ii)	<i>any two from</i> acts as <u>competitive inhibitor</u> ; binds to active site, qualified ; e.g. instead of, substrate / polyprotein to prevent, entry / binding, of, substrate / polyprotein so competes with substrate to prevent ESC formation I <i>ref.</i> to flap region as active site <i>any one from</i> will be permanent / non-reversible ; causes / induces, flap region to open to bind to active site ; no (A few / fewer) polyprotein products / polyprotein not cut, so viral replication prevented / AW ; R polyprotein not broken down	2